TEXORA · LECTURE SHEET

# Chemicals and Auxiliaries Used in Preparatory Wet Processes

*The working chemicals of the preparation department: detergents and surface-active agents classified by charge, the micelle and its critical concentration, the HLB scale, how wetting, detergency and emulsification actually work, and the acids, alkalis, salts, oxidising and reducing agents every recipe is built from.*

TE-2205 · TEXTILE CHEMISTRY & COLORATION-I · TOPIC 3 · LEVEL 2 TERM 2

## 1. Surface-active agents: the general concept

A **surface-active agent** (surfactant) is a molecule built from two opposed parts: a **hydrophobic tail** (a long hydrocarbon chain, typically $C_{12}$–$C_{18}$) that avoids water, and a **hydrophilic head** (an ionic or polar group) that seeks it. Because neither part can be satisfied in the middle of a liquid, surfactant molecules collect at interfaces — air/water, oil/water, fabric/water — and lower the **surface tension** of water from about 72 mN/m towards 30–35 mN/m. That one effect drives almost everything a preparation bath does: the liquor spreads on a hydrophobic fabric instead of beading up (wetting), greasy soil is detached and held in suspension (detergency), and two immiscible liquids can be held as a stable dispersion (emulsification).

## 2. Classification of detergents and surfactants

Surfactants are classified by the charge (if any) on the hydrophilic head in solution:

| CLASS | HEAD GROUP | EXAMPLES | BEHAVIOUR / TEXTILE USE |
|---|---|---|---|
| Anionic | Negative: $–COO^-$, $–OSO_3^-$, $–SO_3^-$ | Soap (sodium stearate), sodium lauryl sulphate, alkylbenzene sulphonates | Main detergents and scouring agents; excellent soil removal; precipitated by hard water (soaps especially) |
| Cationic | Positive: quaternary ammonium $–N^+(CH_3)_3$ | Cetyltrimethylammonium bromide (CTAB), softener bases | Substantive to (negatively charged) fibre surfaces; used as fabric softeners and fixing agents; poor |

| | | | |
|---|---|---|---|
| | | | detergents; incompatible with anionics |
| Non-ionic | No charge: polyoxyethylene – $(OCH_2CH_2)_nOH$ chains | Fatty-alcohol ethoxylates, alkylphenol ethoxylates (being phased out as APEO), Tween series | Unaffected by hardness or pH; good emulsifiers and wetting agents; show a cloud point on heating |
| Amphoteric (zwitterionic) | Both + and − groups | Betaines (cocamidopropyl betaine) | Mild, compatible with all classes; speciality uses |

**Detergents** are formulated products — a surfactant system plus builders (alkali, sequestering agents) and other aids — whereas the surfactant is the active molecule itself. In hard water, anionic soaps fail first because $Ca^{2+}$/$Mg^{2+}$ precipitate them as lime soap (Topic 2); synthetic anionics and non-ionics tolerate hardness far better.

## 3. Micelles and critical micelle concentration

As surfactant concentration rises, the interface becomes saturated; beyond that point added molecules take the only remaining escape route and aggregate in the bulk: the hydrophobic tails hide inside, the hydrophilic heads face the water. This spherical aggregate (≈ 50–100 molecules) is a **micelle**, and the concentration at which micelles begin to form is the **critical micelle concentration (CMC)**. At the CMC, surface tension stops falling and levels off — beyond the CMC, extra surfactant builds micelles instead. Micelles matter practically because oily soil can dissolve into their hydrocarbon cores (**solubilisation**): detergency and emulsifying power only become strong at and above the CMC, so recipes are dosed relative to it.

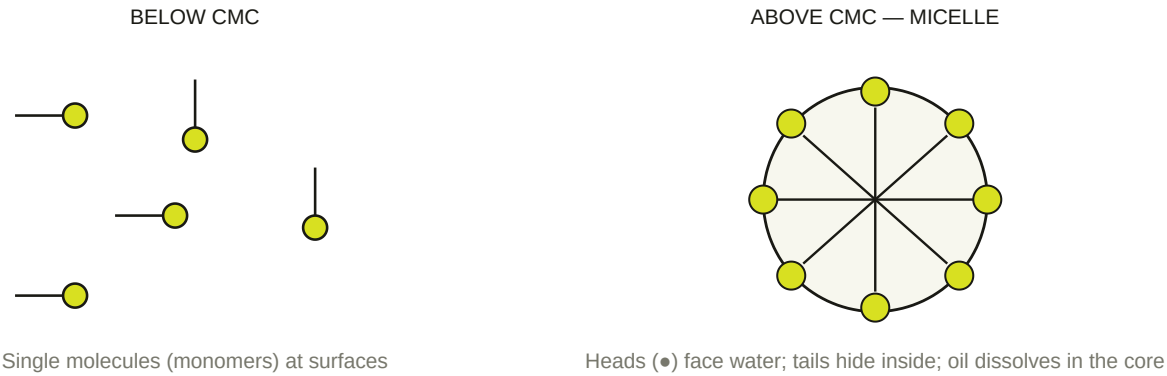



Fig. 3 — Monomers below the CMC; micelle formation above it (schematic)

Fig. 3 — Surfactant behaviour below and above the critical micelle concentration.

## 4. HLB value

The **hydrophile–lipophile balance (HLB)**, on Griffin's scale of roughly 0–20, expresses how water-loving a surfactant (or blend) is, and predicts its job: low HLB stabilises water-in-oil emulsions, high HLB oil-in-water emulsions, detergents and solubilisers.

| HLB RANGE | APPLICATION |
|---|---|
| 3–6 | Water-in-oil (W/O) emulsifier |
| 7–9 | Wetting agent |
| 8–18 | Oil-in-water (O/W) emulsifier |
| 13–15 | Detergent |
| 15–18 | Solubiliser |

$$\text{HLB} = 20 \times (\text{mass of hydrophilic portion}) \div (\text{total molecular mass})$$

**WORKED EXAMPLE**

A non-ionic surfactant has molecular mass 600, of which the polyoxyethylene (hydrophilic) part is 440. HLB = 20 × 440/600 = **14.7** — in the detergent / strong O/W emulsifier range, matching its use as a scouring detergent. Blends are additive by weight: mixing equal weights of HLB 4 and HLB 16 surfactants gives a blend HLB of about 10.

## 5. Mechanisms: wetting, detergency, emulsification

### 5.1 Wetting

A drop on a surface settles at a **contact angle** θ set by the balance of surface energies (Young's equation): $\gamma_S = \gamma_{SL} + \gamma_L \cos\theta$. $\theta > 90°$ means non-wetting (water beads on waxed cotton); $\theta \to 0°$ means complete spreading. Surfactants adsorb at the fabric/liquor interface, lower $\gamma_L$ and $\gamma_{SL}$, and so collapse the contact angle — the liquor floods into yarn interstices and fibre capillaries, expelling air. This is why a wetting agent is the first chemical in almost every preparation bath.

### 5.2 Detergency

Removal of oily/particulate soil proceeds in stages: (1) wetting of fabric and soil; (2) **roll-up** — surfactant adsorption at the oil/fabric interface increases the oil's contact angle until the film contracts into droplets that detach; (3) **emulsification and solubilisation** — detached oil is dispersed as emulsion droplets or dissolved in micelles; (4) **suspension** — soil is kept dispersed (charge repulsion, builders) so it rinses away instead of redepositing. Insoluble-solid soil is removed by the same wetting plus mechanical action and anti-redeposition agents such as CMC.

### 5.3 Emulsification

An **emulsion** is a dispersion of one immiscible liquid in another (oil-in-water, O/W, or water-in-oil, W/O). A surfactant film at each droplet surface lowers interfacial tension and (for ionics) gives droplets like charges that repel, preventing coalescence. Scouring works because cotton wax is emulsified into the bath; spinning-oil removal from synthetics is emulsion scouring; and batching oil on jute is applied *as* an emulsion.

## 6. Acids, alkalis, salts, oxidising and reducing agents

| FAMILY | MEMBERS | PROPERTIES AND USES IN WET PROCESSING |
|---|---|---|
| Alkalis | Caustic soda NaOH (strong), soda ash $Na_2CO_3$ (mild), ammonia, trisodium phosphate | Scouring cotton (saponifies wax), mercerisation (conc. NaOH), pH control, wool/silk handled only with mild alkali |
| Acids | Acetic and formic (weak, common), sulphuric (strong), hydrochloric | Neutralising alkali after scouring, pH setting for dyeing, carbonising vegetable matter in wool ($H_2SO_4$), acid desizing |
| Salts | Common salt NaCl, Glauber's salt $Na_2SO_4 \cdot 10H_2O$, sodium silicate | Electrolyte to promote dye exhaustion (dyeing), silicate as peroxide stabiliser, water treatment |
| Oxidising agents | $H_2O_2$, NaOCl, $NaClO_2$, $KMnO_4$, persulphates | Bleaching (destroy chromophores), oxidative desizing, discharge printing |
| Reducing agents | Sodium hydrosulphite (dithionite), thiourea dioxide, sodium sulphide, glucose | Vat and sulphur dye application (reduce dye to soluble leuco form), reduction clearing, stripping faulty dyeings |
| Other auxiliaries | Sequestering agents (EDTA, phosphonates), carriers, levelling agents, enzymes | Bind metal ions, assist disperse dyeing, equalise dye uptake, specific mild preparation (Topic 1) |

## 7. Glossary — key terms

- **Surfactant:** a molecule with a hydrophilic head and hydrophobic tail that collects at interfaces and lowers surface tension.
- **Micelle:** a spherical aggregate of surfactant molecules formed above the CMC, tails inward.
- **CMC:** the concentration at which micelles begin to form and surface tension levels off.
- **HLB:** hydrophile–lipophile balance; 0–20 scale predicting emulsifier/detergent character.

- **Emulsion:** a stabilised dispersion of one immiscible liquid in another (O/W or W/O).

## 8. Key points

- Surfactants are classified by head-group charge: anionic (main detergents), cationic (softeners/fixers), non-ionic (hardness-proof emulsifiers/wetters), amphoteric.
- Above the CMC, micelles form and detergency/solubilisation become effective.
- HLB predicts function: low = W/O emulsifier, mid = wetting, high = O/W emulsifier, detergent, solubiliser.
- Wetting is contact-angle collapse; detergency is wetting + roll-up + emulsification/solubilisation + suspension.
- Alkalis scour, acids neutralise and carbonise, salts exhaust dye, oxidants bleach, reductants solubilise vat/sulphur dyes.

## 9. Summary

Preparation chemistry is surfactant chemistry plus a short list of commodity reagents. Surfactants lower surface tension and organise into micelles above the CMC, giving the wetting, detergency and emulsification that cleaning depends on; their HLB value predicts which job each will do best. Around them, alkalis do the heavy chemical work of scouring, acids neutralise and adjust pH, salts control dyeing, and oxidising and reducing agents handle bleaching and the special dye classes.

## 10. Practice

Answer in your own words; brief model answers follow.

1. Why does adding more detergent beyond the CMC not lower surface tension further, yet still improve oily-soil removal?
2. A surfactant of molecular mass 500 has a hydrophilic portion of 300. Calculate its HLB and name a suitable use.
3. Explain roll-up in the detergency mechanism, and state what happens to the detached oil afterwards.
4. Sodium hydrosulphite and hydrogen peroxide are both found in a dyehouse. State the function of each and the class of process it belongs to.
5. A scouring bath made with soap performs badly when the mill switches to harder well water, but a synthetic anionic/non-ionic detergent is unaffected. Explain.

### Answer key

1. Once the interface is saturated (at the CMC) extra molecules cannot reduce tension further; they form micelles instead. Oily soil dissolves into micelle cores (solubilisation) and micelles also stabilise emulsified oil, so cleaning still improves.
2. HLB = 20 × 300/500 = 12 — an O/W emulsifier suitable as a general emulsifying/washing auxiliary (just below the typical detergent band).
3. Surfactant adsorbs at the oil–fabric interface and lowers interfacial tensions so the oil film's contact angle increases; the film contracts into droplets and detaches. The droplets are then emulsified or solubilised into micelles and kept suspended until rinsing carries them away.
4. Sodium hydrosulphite is a reducing agent — it reduces vat/sulphur dyes to their soluble leuco forms and is used for reduction clearing and stripping. Hydrogen peroxide is an oxidising agent — it destroys chromophores in bleaching (and is used in oxidative desizing).
5. $Ca^{2+}$/$Mg^{2+}$ precipitate soap as insoluble lime soap, removing the active surfactant and depositing curd. Synthetic anionics form more soluble Ca salts and non-ionics carry no charge at all, so neither is precipitated by hardness.

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